PrestoID — Product & Technical Specification

A Coaching, School, Library & Institution Management Platform

By Team43

Version 1.0 — Prepared for development handoff (Antigravity IDE)

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1. Product Overview

PrestoID is a smart coaching/institution management app built for small-city coaching centers (NEET, MPPSC, SSC, Banking, Railway, UPSC, etc.). It has two sides of one app:

- **Coaching Panel** — for the institute owner / teacher / admin
- **Student Panel** — for enrolled students

A student can be enrolled in **multiple coachings** at once (e.g., Physics coaching + separate Biology coaching), with one login but switchable coaching profiles.

The product is separate from Kanelijo. Branding: **"PrestoID by Team43."**

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2. App Navigation Structure (Final)

2.1 Student Bottom Navigation

Tab	Purpose
ID Card	Digital virtual ID, QR code for attendance
Community	Feed of posts/notes/announcements from coaching
Test	AI-generated tests, schedule, history, analytics
Alerts	Notifications (fees, attendance, community, test schedule)
Profile	Personal info + Attendance + Fees (via pill-tabs)

Profile screen — internal pill-tabs:

- Personal Details
- Attendance
- Fees

2.2 Teacher / Coaching Bottom Navigation

Tab	Purpose
Students	Manage all registered students, create profiles, QR scan
Community	Post announcements, notes, files to students
Test	Create AI-generated tests, manage question bank, view results
Alerts	Notifications + pending claim approvals
Profile	Coaching info, batches, settings, notification toggles

Students panel — internal structure:

- Student list (search, filter by batch)
- "Fee Overview" toggle/view → global collected/pending dashboard (replaces separate Fees tab)
- Tap a student → Student Detail page with pill-tabs:
  - Profile
  - Fees
  - Attendance
  - Test Analysis

> **Note:** The standalone "Fees" tab is removed from both bottom navs. All fee data now lives inside Profile (student side) and inside Students → Student Detail / Fee Overview (teacher side). The UI component pattern reuses the existing pill-tab toggle (already built for Alerts/Community Feed switching).

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3. Multi-Coaching Account Model

- **Single app login** (Google Sign-In), shared across all coaching memberships.
- A student can belong to multiple coachings — each is a separate "coaching profile" attached to the same account.
- **Switching profiles:**
  - On app open: if multiple coaching profiles exist, prompt to select one.
  - From Profile section: always available to switch.
- **Joining a second (or first) coaching:** Student selects coaching from a dropdown list, then must complete the claim process (see Section 4).

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4. Student Onboarding & Profile Claim Flow

4.1 Teacher Creates Profile First (Mandatory)

- Students **cannot self-register**. A teacher must first create the student's profile via the "+" button in the Students panel.
- Teacher fills in at minimum:
  - Full Name
  - Batch
  - Aadhaar Number
  - WhatsApp Number
  - Date of Joining / Validity period
- On **Save**:
  - System auto-generates a **unique Secret Code / Enrollment ID** (see Section 4.3)
  - Code is automatically sent to the student's saved WhatsApp number

4.2 Student Claims Profile — Two Methods

Method	Flow
Secret Code (Primary)	Student enters the code received via WhatsApp → instantly linked to profile
Aadhaar Number (Backup)	Student enters Aadhaar number → claim request enters **Pending Approval Queue** in teacher's Alerts panel → teacher reviews (sees matched name/photo) → **Approve** or **Decline**

> **Critical safeguard:** On Aadhaar-based claims, the teacher's approval screen must clearly display the matched student record's name and photo (if available) to prevent mismatched/incorrect approvals.

4.3 Unique ID / Secret Code Format

Format: `[INSTITUTE_CODE]-[YEAR]-[4_DIGIT_ID]`

Example: `UCI-2026-1936`

- **INSTITUTE_CODE**: 3–4 letter abbreviation chosen by coaching at signup; must be globally unique across the platform.
- **YEAR**: Year of enrollment/joining.
- **4_DIGIT_ID**: Auto-incrementing per institute (1001, 1002, 1003...).

This single ID serves **three purposes**:

1. Secret claim code (sent via WhatsApp)
2. Visible Enrollment ID on the digital ID card
3. QR code payload (scanned by teacher for attendance)

4.4 Until Claimed

- A teacher-created but unclaimed profile remains inactive/locked from the student side.
- Profile cannot be claimed for a second coaching until that coaching's teacher has created the student's profile there too — the claim flow is identical and independent per coaching.

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5. Profile Completeness & Sync

5.1 Skeleton Profile → Student Completes the Rest

- Teacher-entered fields (name, batch, Aadhaar, WhatsApp, validity) are the **baseline**.
- All other fields (photo, DOB, blood group, father's name, father's phone, home address, etc.) start empty.
- Student's Profile screen shows a **completeness indicator** (e.g., progress bar / checklist badges — see "Name ✓ DOB ✓ Phone ✓ ... Photo ✓" tag row from reference screenshots).
- Empty fields render as "tap to add" prompts.

5.2 Real-Time Sync (Supabase Realtime)

- **Student record changes** (teacher edits name/batch/validity, etc.) → pushed live to student's app via Supabase Realtime subscriptions (Postgres change events over websocket). No manual refresh needed.
- **Community feed** → new posts pushed live to all subscribed students in that coaching.
- Tables requiring realtime subscriptions: `students` (own record), `posts`, `alerts/notifications`, `attendance_status`.
- Tables NOT requiring realtime (fetch-on-load is sufficient): `fee_history`, `test_results`.

5.3 Offline Behavior

Scenario	Behavior
Offline, cached data exists	Show last-synced data with a subtle "Offline — showing data from [time]" banner. No blank screens.
Offline, no cached data	Explicit empty-state: "No internet connection — connect to load your dashboard" + retry button. Never show misleading empty lists.
Connection restored	Auto-revalidate, re-subscribe to realtime channels, brief "Back online — syncing..." toast, flush any queued actions.

5.4 Offline Write Actions

Action	Offline Behavior
QR Attendance Scan	**Queued locally** with timestamp → synced to server on reconnect (shown as "pending sync" in teacher UI)
Fee marking, community posts, profile edits	**Require live connection** — fail with "No internet — try again" if offline

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6. Digital ID Card & QR Attendance

6.1 ID Card Contents

- Coaching/Institute name + logo
- Student photo, name, Enrollment ID (`UCI-2026-1936`)
- Course/Batch name, Batch timing, Duration
- Valid From / Valid Till dates
- QR code (back of card / flip) — encodes the Enrollment ID
- "Flash for Attendance" action button
- Live status banner showing current batch session if "live" (e.g., "● LIVE AT UCI — MPPSC — 10:00 AM—01:00 PM — Room 3")

6.2 Attendance Marking

- Teacher scans student's QR (via "Scan QR Code" in Students panel) → attendance is recorded **with timestamp**.
- **Attendance is general** — any scan at any time logs a presence record with that exact timestamp. No requirement to "select batch first."

6.3 Auto Batch-Tagging by Time

Each batch has a configurable time range (set per-coaching, not hardcoded). The scan timestamp is matched against batch time ranges to auto-assign which batch session the attendance belongs to.

Example default bands (customizable per coaching):

Scan Time	Auto-Tagged Batch
Before 9:59 AM	Morning
10:00 AM – 1:59 PM	Afternoon
2:00 PM onward	Evening

- If scan time falls outside all defined batch ranges, or overlaps multiple batches → tagged as **"General"**, with manual reassignment available to the teacher from the attendance log.

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7. Community Feed

7.1 Structure (v1 — No Targeting Algorithm)

- **Single common feed** visible to all registered students of a coaching.
- Batch-specific targeting is explicitly **deferred to a later version**.

7.2 Post Types & Content

- **Post categories:** Announcement, Note (as seen in reference UI — color-coded tags)
- **Content types supported:** PDF, Image, Text (with carousel/multi-image support). **No video** in v1.

7.3 Engagement Features

- Like
- Comment (+ comment-on-comment / threaded replies)
- Download (for PDFs/files)
- "Viewed by" list (Instagram-style)

7.4 Feed Loading & History

- **Cursor-based pagination** — loads ~10–15 posts at a time, infinite scroll.
- **30-day server-side history window** — older posts archived/not returned by default to maintain query performance. Beyond-30-days access is a future enhancement (search/load-more).

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8. AI-Powered Test System

8.1 Test Bank Workflow

1. **Content Upload:** Teacher uploads syllabus materials — **PDF, image, or text** — building a permanent, reusable **Test Bank** per subject/chapter.
2. **Test Creation Request:** Teacher specifies parameters in natural language / structured form:
   - Batch (by name or time)
   - Test duration (e.g., 60 minutes)
   - Topic/chapter range (e.g., "Indus Valley Civilization to Mahajanapadas")
3. **AI Question Generation:**
   - AI reads uploaded materials, extracts content with **high accuracy**
   - Generates MCQs with 4 options + correct answer + solution/explanation
   - **Each question is tagged with chapter/topic** for later analytics (e.g., "Indus Valley Civilization")

8.2 Teacher Review (Mandatory Before Go-Live)

- **Every AI-generated question must pass through a teacher review screen** before the test goes live.
- Teacher can edit question text, options, correct answer, or explanation for any question.
- No AI-generated test publishes directly without this review step.

8.3 Scheduling & Notification

- Teacher selects target batch (by name or time).
- All students whose profile matches that batch **automatically receive a notification**: "Your [Subject] test is scheduled at [time]."
- Tapping the notification opens full test instructions: duration, topic range, start time.

8.4 Test-Taking Experience (Local-First / Offline-Capable)

This is the most critical architectural decision in the system — **the test runs entirely on-device once started.**

At test start:

- Client downloads: all questions, all options, full duration, **server-issued start timestamp**.
- From this point, the test does **not require network connectivity** to function.

During the test:

- Local countdown timer, anchored to `serverStartTime + duration`, validated against a **monotonic clock** (not wall-clock `Date.now()`) — prevents cheating via device clock manipulation.
- Answers save **incrementally to local storage** (not network calls).
- **Connection loss has NO effect on the test** — it continues uninterrupted.

Exit / App-Switch Detection (separate from connectivity):

- If student switches away from the test app (minimizes, opens another app):
  - **20–30 second grace timer** starts
  - **Exit counter increments** (visible to teacher in analytics)
  - If student returns within grace window → test resumes normally
  - If not → **auto-submit** with answers saved so far

Submission:

Scenario	Behavior
Online at submit time	Immediate upload → AI grading → analytics generated
Offline at submit time	Local "pending submission" flag → student sees "Test completed — will submit once you're back online" → auto-uploads on reconnect
Timer expires while offline	Device enforces cutoff locally (via monotonic clock) → auto-submit → marked "pending submission" → uploads on reconnect

8.5 Auto-Grading & Analytics

- AI matches submitted answers against correct answers — fully automated, "middleman" grading (no manual teacher grading needed).
- **Student-side:** receives summary + analytics immediately after submission (score, chapter-wise breakdown — e.g., "Weak in: Indus Valley (40%) | Strong in: Mahajanapadas (85%)").
- **Teacher-side:** full report appears in the relevant student's **Test Analysis** tab, including:
  - Score, rank (if applicable)
  - Chapter-wise weak/strong areas
  - **Exit log**: number of exits, total time away, and exit type per event (`app_switched` vs `connection_lost` — logged separately for context, even though both are handled by the same timer/counter mechanism)

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9. Fees Management

9.1 Student Side (within Profile → Fees tab)

- Current Due (amount, month, due date, days remaining)
- Total Paid / Total Pending summary
- Payment History
- Informational note: *"Fees are collected by your coaching institute. Contact your admin for payment methods, receipts, or fee-related queries."*

9.2 Teacher Side (within Students → Fee Overview)

- Global dashboard: % collected, Total Collected (₹), Total Pending (₹)
- Filter by month
- Filter tabs: All / Paid / Unpaid / Overdue
- Per-student fee status visible in list
- "Export Report" and "Remind All Unpaid" actions

9.3 Reminders (Phased Rollout)

Phase	Channel
Phase 1	In-app notification only
Phase 2	WhatsApp reminders (Auto Fee Reminder: 15th & last day of month)
Phase 3	In-app payments via Razorpay

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10. Single-Device Login & Security

10.1 Login Enforcement

- One student account = **one active device** at a time.
- New login attempt from a different device:
  1. **Approval request sent to the previously logged-in device** — high-priority push notification with sound, presenting **"Was Me" / "Wasn't Me"** action buttons.
  2. New device **waits** for response before being granted access.
  3. **If approved ("Was Me")** → old device logs out, new device gains access.
  4. **If declined ("Wasn't Me")** → new login is denied.
  5. **If no response within timeout window** (e.g., 10 minutes) → **auto-deny** (fail-safe default) → new device sees: "Approval timed out — contact your coaching to reset device access."

10.2 Recovery Path — Teacher Device Reset

- Covers: lost/dead/sold old device, or old device unreachable.
- Teacher has a **"Reset Device Lock"** button on each student's profile (Students → Student Detail → Profile tab).
- Clears the stored `device_id` — next login from any device succeeds without requiring approval.
- This is the **single unlock mechanism** for both "lost device" and "unreachable device" scenarios.

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11. Aadhaar Data Handling — Compliance Note

- Aadhaar numbers are sensitive government ID data and **must be encrypted at rest** (column-level encryption in Supabase) per India's DPDP Act 2023 requirements.
- Aadhaar is used only as a **backup claim method** (primary = secret code via WhatsApp).
- All Aadhaar-based claims require explicit teacher approval (see Section 4.2).

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12. Notification Settings (Teacher-Configurable, per Coaching)

Setting	Default	Behavior
Auto Absent Alert	ON	WhatsApp to parents at 8 PM daily
Auto Fee Reminder	ON	WhatsApp on 15th & last day of month
Community Notifications	ON	Notify students on new community activity

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13. Free Trial & Subscription Model

13.1 Trial Terms

- **7-day free trial** with access to **all paid features**.
- After trial expiry, paid features lock; free features remain accessible **indefinitely**.

13.2 Feature Tiering

Free (Always Available)	Paid (Locked After Trial)
Students panel (manage, create profiles)	AI Test Generation
Community (post, feed, like/comment)	WhatsApp Reminders (fee + absent alerts)
ID Card / QR Attendance	Razorpay In-App Payments
Alerts (in-app notifications)	
Fees Tracker (manual mark paid/unpaid)	

Rationale: Free-tier features have near-zero marginal cost to PrestoID (database operations only). Paid-tier features carry real per-use costs (LLM tokens for AI test generation, WhatsApp Business API per-message fees, Razorpay transaction fees). This structure keeps non-converting coachings active on the platform using free features — building habit and serving as a funnel toward conversion via the AI test-generation "hook."

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14. Settings (Profile Tab — Both Sides)

Student Settings

- Notifications: Attendance Alerts, Fee Reminders (toggles)
- Account & Privacy: Change Password, Help & Support, Terms & Privacy Policy
- Danger Zone: Log Out, Delete Account
- Footer: App version + "by Kanelijo" / "by Team43" branding

Teacher Settings (within Profile tab)

- Coaching Center info: Institute Name, Invite Code, Location
- Batches management: Add/edit batches (with time-range configuration for attendance auto-tagging)
- Notification Settings (Section 12)
- Account: Change Password
- "Reset Device Lock" — accessible per-student from Student Detail view

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15. Open Items for Future Phases

- Batch-specific community feed targeting (algorithm/tagging-based)
- WhatsApp-based fee/absence reminders (Phase 2)
- Razorpay in-app payment integration (Phase 3)
- Feed history beyond 30 days (search/archive access)
- Multi-language support for AI-generated test content

End of Specification — PrestoID v1.0 by Team43